TBA 414

Topic: Maintenance and Replacement model (Analysis). Prof. Ogbechi

Replacement: This is act of substituting a bad thing with a good thing replacing something defective.

Maintenance is the act of making things last longer. Maintenance is used when the thing involved is good, you are maintaining it, rather than replacing it.

Organisation is made up of social technical components (human and Non-human component). These components must be operational for the business to function properly.

* Understanding:

Replacement Analysis can be simply viewed as the process by which the various cost consequences involved in replacement are studied so that optimal replacement decision are taken. for you to replace or maintain, you must study the cost implication implementation.

Types of Items in Organisation.

The organisation is a socio-technical context that is made up of different component that interact & interrelate in order to achieve organisational goals. all these components have to be functioning properly at all point in time, but for these to be achieved they need to be replaced or maintained. the question then is, what do we have to replace and what do we have to maintain. To answer these questions, a knowledge of

Types of Items in org is of essence:

A) The Items in orgs can be classified into three:

1- Sudden failure Items: these are items working perfecting but stops suddenly without any prior notice eg fan, bulbs. the consequential cost of failure can be substantial but the cost of the item itself might be cheap. high

2.) Deteriorating items: These items deteriorate with time and usage. These items that wear and tear as we keep on using them. As time passes the depreciate. These items are bulky, expensive and cannot be replaced so it is better to be properly maintained eg generators, machine, buildings, vehicles etc.

3.) Depreciating items: These are items that depreciate in efficiency with the passage of item until they fail. (maintenance only increase it life span Note: for the deteriorating items proper maintainance enables the item to last for ever. (eg proper maintenance of a house can make it last for 500 years While depreciating items even with maintenance, it does not last forever Note: for deteriorating items when the maintenance cost is too high, you just have to let go and replace the item.

B) Assumptions of Replacement Analysis

1. Objective is to minimise cost.
2. Technological level is constant.
3. Replacement cost is constant.
4. Maintenance cost is constant.
5. Output quality is constant.
6. Performance level is constant.

Replacement models.

A. Sudden failure items: for us to apply replacement models in these we need to know the various cost.

i) Cost involved are as followed:

a) The replacement cost of the item (purchase price of the item - off item)

b) The consequential cost of failure (e.g lost of goodwill, idle time, Labour frustration, Penalty cost (these cost are usually substantial

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sf of Replacement.: The actual cost involved in replacing the item. It is
2 totality of all the other cost plus the labour (charge) cost. i-e The replacement
:st + Consequential cost + labour charges.

Replacement policies for sudden failure items.

ere are two types:

Individual replacement policy: The replacement is done individually
:the items fails e.g if we have ten bulbs in a room, if one
ils, it will be replaced immediately.

Group replacement policy: This policy is applicable when the
rganisation choose a particular point in time either annually, wee
y monthly etc when replacement will be made both on defective
nd functional items.

ote: As a manager, you consider the cost and compare btw the two
wo model which are cheaper as we have to use "algorithm."

llustration: The operations manager of Danyan nall, a specialised eye
lasses manufacturing outfit is considering the optimal replacement policy
) H
rA special translucent light bulbs in the assembly line. The individ
dual replacement cost per bulb as it fails is ₦5000 while the
ost of replacing the bulbs as a group is ₦20,000. Assume that
re initial number of translucent light bulbs in the assembly line is 200
ulbs. The table below gives the failure distribution frequency
er period.

eriod of failure (t)	Frequency of failure: Period of failure in years
1	3
2	20
3	15
4	12
5	30

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required: (a) Advise the operations manager on the best policy to a

Mechanics of individual replacement policy

steps)
1. find the probability of failure per period for the items.
2. finel the average life of the item (AL). This is equal to the summation of the probability of the item multiplied by the time of the period.

$$AL = \sum P_i t_i$$

P -> Probability for items, Period
Where: t -> time period for items, Period

3. find the number of items to be replaced: N_r < No of items to be replaced

$$N_r = \frac{No}{AL}$$

Where: No -> Initial number of items.
A_L -> Average life.

4. Find the Individual replacement cost. (I_{rc})

$$I_{rc} = N_r \times P$$

N_r -> No of items to be replaced
Wher P -> purchase price of item.

solution.

Period of failure (t)	freq of failure	Prob of failure (P_i)	Average Life AL $\sum P_i t_i$
1	3	3/80 = 0.04	0.04
2	20	0.25	0.50
3	15	0.19	0.57
4	12	0.15	0.60
5	30	0.37	1.85
	80	1.00	AL = 3.56

step3: $N_r = \frac{No}{AL} = \frac{200}{3.56} = 56.1 \simeq 56.$

step4: $I_{rc} = N_r \times P = 56 \times ₦5000 = ₦280,000.$

Group Replacement Policies for Sudden failure Items.

Mechanics for group replacement for sudden failure items.

step1: tabulate the time period column wise.

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tep2: find the probability of failure
-ep3: find the number to be replaced per period for the Item
The number to be replaced per period is obtained using the following
ormular. (N_r).

$N_1 = N_0 P_1 = (200 \times 0.04) = 8$.

$N_2 = N_0 P_2 + N_1 P_1 = 200(0.25) + 8(0.04) = 50 + 0.32 = 50.32 \simeq 50$

$N_3 = N_0 P_3 + N_1 P_2 + N_2 P_1 = 200(0.19) + 8(0.25) + 50(0.04) = 42$.

$N_4 = N_0 P_4 + N_1 P_3 + N_2 P_2 + N_3 P_1 = 200(0.15) + 8(0.19) + 50(0.25) + 42(0.04) = 46$

$N_5 = N_0 P_5 + N_1 P_4 + N_2 P_3 + N_3 P_2 + N_4 P_1 = 97$

ep 4: Find the cummulative number to be replaced by period.
ep5: find the cummulative individual cost per period (this is
qual to cummumlative number to be replaced by period multiplied
y the Individual replacement cost per unit.
tep 6: Find the fixed cost (this is equal to the Initial cost of
th Item (No) X the group replacement cost given.
tep7: find the total cummulative replacement cost = Addition of
ummulative Individual replacement cost and the fixed cost per period.
ep8: find the average total cummulative replacement cost = Total
ummulative replacement cost per period ÷ respective time period.
ep9: Select the least average total cummulative replacement cost
nd conclude that group replacement should be done at the end
=that period.
ep 10: If you are told to compare the Individual replacement policy
placement period, then you compare the selected value in step 9
th your computed Individual replacement cost value then select the
ast cost policy.
ep 11: Stop.

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Period of failure (t)	Freq of failure	Prob of failure	N_t	Cum N_t	Ind cost = 5000 Cum Ind cost	$N_0 \times G_c$ FC = 200 × 20,000	TCRc (6+7)	ATCRc 8/1
1	3	0·04	8	8	8×5000 = 40,000	4,000,000	4,040,000	4,040,000/1 = 4
2	20	0·25	50	58	58×5000 = 290,000	4,000,000	4,290,000	2,145,00
3	15	0·19	42	100	100×5000 = 500,000	4,000,000	4,500,000	1,500,000
4	12	0·15	46	146	730,000	4,000,000	4,730,000	1,182,50
5	30	0·37	97	243	1,215,000	4,000,000	5,215,000	1,043,00
	80	1·00						

From the cal. above I advise the company to use th individual replacement method rather than a gro

$N_5 = 200(0.37) + 8(0.15) + 50(0.19) + 42(0.25) + 46(0.04) = 97$ "

Maintenance of detriorating Items:

When dealing with detrioating Items we are faced with major cost namely:
i) Increasing maintenance cost. ii) Decreasing value of the item with the passage of time. eantiy

The depreciation is the difference btw the market value at the beginning of t year and at the end of the year. The decision rule is to determine t time period when the combination of this maintenance policy cost a at a minimal. several models can be used to determine this, ho -ever, we shall be applying the differential calculus method or the mini average total cost ~~(inventory)~~ of the solar system at the

Example: The annual average cummulate cost (y) of the solar system at the fa -ty of management system class room & expressed a function of its life span (x) is given by the equation. $y = 10x^2 + 800x + 5000$. Required: Determine the optimal replacement age of the solar inverter syst

solution.

Recall that differentia equation formulat

$dy/dx = nCx^{n-1}$.

Where n → is the power to which x is raised.

C → is the constant

x → Variable in this case tim

for the first equation $n=2$; $C=10$

$\therefore \frac{dy}{dx} = 2(10)x^{2-1} + 1(800)x^{1-1} + 0$

$dy/dx = 20x + 800x^0$

$dy/dx = 20x + 800$

At turning point $dy/dx = 0$

$\Rightarrow 0 = 20x - 800$

$20x = 800$

$x = 800/20 = 40$.

$x = 40$ years.

Note: The two methods that can be used under maintenance policy model for deteriorating and depreciating items are;
i) The minimum average total cost method; This metho is based on decision theory.
ii) The differential calculus method; Th is based on differential calculus.

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Maintenance of detriorating items:

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Example: The annual average cummulate cost (y) of the solar system at the facu-ty of management system class room is expressed a function of its life span (x) is given by the equation. $y = 10x^2 + 800x + 5$ Pou.
Required: Determine the optimal replacement age of the solar inverter syste

solution.

formular

Recall that differentia equation $dy/dx = nCx^{n-1}$.

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x → variable in this case time

for the first equation $n = 2$; $C = 10$

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Topic: Linear Programming.

05/01/22. 5/01/22.

inear programming is a mathematical model use for allocating scare resources
o uses in an optimal manner. The model explore mathematical symbo
ls and expression in modelling a problem situation subject to the
vailability of resources. It is a uni-objective optimisization model
that trys to either maximise a given function in a profit related
roblem or minimise a given function in a cost related problem.

However, since the maximisation of profit or minimisation of
ost is always a constraint by real life resource limitation. The objective
unction in a linear program is constraint by real world resou
ce avalability.

Areas of application of linear programming.

The areas of application of linear programming includes, but
of limited to, the following:

- Investment planning: That is deciding how much to Invest in
ch Investment alternative. Investment can be in stock or
onds.

2. Transportation Arrangement: That is deciding what mode of trans
portation to be used to ship goods. production planning
:) Production planning: That is deciding on what product
o be produced at what time
) production scheduling: That is deciding on what product
o be produced on which machine and in what order
) co-operate planning: That is, using linearing programming to
nap out the strategy of a whole companies operation.

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6) Advertising and promotion campaign: That is, using linear prog to decide or select which media should be used for advertising promotion. The purpose of advertising is to get many people to informed at a particular time. Print Media eg News paper, electron media eg Radio, television etc.

7) State security: Linear programming can be used to map out strat military effectiveness.

8) Crime prevention and control: L.P can be used to determine the opt police personnel that is required to prevent and control crime in a

Terms Used in Linear Programming (Terminologies).

1. Decision variables: These are variables or unknown quantities, repre ted in a linear programming model by the mathematical symbol. " X_j or X_{ij} ", whose values will seek to determine.

2) Contribution or (cost) coefficient: These are constants that expresses the per Unit contribution (or cost) of decision variable X_j to the overall or total profit (or cost). It is denoted by symbol (C_j) eg There is o one company that can be laid with thermo-cool refrigeration and it only P2 industry that produces the refrigeration bcos the P2 company knows that it have comparative advantage, the decided to produce various sizes. It u the material required, same processes but different sizes are available but P2 cannot decide what will give them the overall total profit for the p Let X_j (J = 1, 2, 3, 4) denote the number of units to product j to produce in a year. Note! the profit you make for selling a particular units will be the same for each size eg Profit made on unit (1) = C_1X_1 while for un 2 is C_2X_2. while for Unit 3 is C_3X_3 and Unit 4 is C_4X_4 et if $Z = C_1X_1 + C_2X_2 + C_3X_3 + C_4X_4$

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Technological Coefficient: Technological coefficient, denoted by the mathematical symbol "a_{ij}" is the quantity of resource i that goes into the production of one unit of resource or product j

e.g. R_1 → Material 1 ; R_2 → Material 2 ; R_3 → Process 1 ; R_4 → Process 2

Suppose we have two material and two processes required to produce thermo cool refrigerator. Note P2 limited is limited in resources & process

Subject to: $a_{11}x_1 + a_{11}x_2 + a_{13}x_3 + a_{14}x_4$ (i.e)

$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4$

$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4$.

$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4$.

i) Right-hand side value: The right hand side value in a L.P, denoted by b_i expresses the amount of resource i that is available in the complex situation. The value is so named by its present at the right hand side of the constrained expression. (b_1, b_2, b_3, b_4).

$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 \leq b_1$

$a_{21}x_1 + a_{22}x_2 + a_{23}x_3 + a_{24}x_4 \leq b_2$

$a_{31}x_1 + a_{32}x_2 + a_{33}x_3 + a_{34}x_4 \leq b_3$

$a_{41}x_1 + a_{42}x_2 + a_{43}x_3 + a_{44}x_4 \leq b_4$.

ii. Slack variable: Slack variable represented by S_i in a linear.P. denote the amount by which the sum of the terms on the left hand side of a constraint expression fall short of the values on the right hand side

Surplus variable : Surplus variable denoted by S_i denotes the amount by which the sum of the the terms on the left hand side of a constraint expression exceeds the values on the right hand side

Eg for slack variable: $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 + S_1 = b_1$.

suppose the constraint is : $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 \geq b_1$.

then we balance the equation: $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 = b_1 + S_1$

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Since S_1 is a variable and not a constant then you re-write the eq

$a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 - S_1 = b_1$ → surplus variable.

while $a_{11}x_1 + a_{12}x_2 + a_{13}x_3 + a_{14}x_4 + S_1 = b_1$ → slack variable <+

Assumptions governing the use of linear programming.

1. Linearity Assumptions: This assumption expects the decision variables a -ing in both the objective functions and the constraint expressions to raised to power one only. further more, the assumption expects ~~to Aon of the~~ the objective function to be single (linear) that is the decision ma has just one objective to achieve.

2.) Addictivity Assumptions: This assumption expects the terms appearing in objective functions and each of the constraint expression to be addict That is to say, the decision variables are independent of one anoth in otherwords, cross products of decision variables such as X_1, X_2 does not exist

3) Divisibility Assumptions: This refers to the fact that decision variables cannot be limited to integer values. In other words decision variabl can take ~~over~~ up integer and fractional values.

4) Deterministic Assumption: This requires ~~that~~ the parameter inputs (ie c_j, a_{ij},) into a linear program should be known with certainty. Thus reflecting de -ion making in a ~~certain~~ static time frame.

Non-Negativity Assumption: This concern the fact that decision variables can take up or have a negative value (either/@ $X_j \geq 0$

5) Non Negativity assumption: Decision variables cannot take on a negg value: ~~part~~

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01/22. FBA 414. DR Abiola.

Topic: Sequencing and Routine. (SEQUENCING)

Every organisation will want to utilise its productive system effectively & efficiently in other to maximise profit by meeting delivery deadlines. Thus ob Involving many operations have to be performed and whereas here are limited resources in terms of plant and machinery on which e job has to be performed.

efinition: Sequencing is concern with selecting the appropriate order r a series of jobs to be done on a finite number of service facilities.

Sequencing based on assumption (Assumptions of sequencing).

All Machines can possess only one job at atime

· Processing time is known and fixed.

· When a job start it must be completed

Receiving Operatings must be completed before the succeding one

· All jobs are arranged in a specified order

All jobs are ready for processing.

All machines have different capability and capacity

· Processing time of a job has no relation in order in which the b is processed

No time is wasted in shifting a job from one machine to another

Types of sequencing problem

ere are three types of sequencing problem:

Type 1 : n Jobs on one machine.

Type II : n Jobs on two machines.

Type III : n jobs on m machines (M>2).

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Terminologies of sequencing.

1. Machine: It is the facility which has some processing capability.
2. Scheduling: This is when time and sequence of processing a job is spe
3. Total elapsed time: This is the time that lapsed between the startin the first job and the finishing of the last job.
4. Idle time: It is the time the facilities or the machines where n used during the total elapsed time.
5. No passing Rule: It refers to the rule of maintaining the order which jobs are to be processed on a given machine ie a p -cular job must first be Completed before anyother one can

Example: The processing time for jobs A, B, C, D and E and the due dates for these jobs in the machine shops are given in the below.

Table 1:

JOB	Process Time	Due Date
A	9	16
B	7	20
C	5	25
D	8	20
E	6	40

Required: The jobs are to be sequenced according to:
i) first come first serve (FCFS) ii) SPT (Shortest processing t
iii) Least Slack (LS) iv) EDD (Expected due date).

solution.

First come first serve (FCFS).

Step 1: Repeat the first two columns.

Jobs	process time
A	9
B	7
C	5
D	8
E	6

p2: cummulate the process time and ld them together <this is called the ow time.

Jobs	Process Time	Flow Time
A	9	9
B	7	16
C	5	21
D	8	29
E	6	35
Cummulative		110

tep3: Add the column of the due date, step 3.

Jobs	Process time	Flow time	Due date
A	9	9	16
B	7	16	20
C	5	21	25
D	8	29	20
E	6	35	40

tep 4: Determine the late job by checking comparing the due date against he flow time Eg JOb A was finished in 9 days against

the promise delivery of 16 days therefore there is no late days or delay

Job B was completed in 16 days against delevary date of 20 days therefore no late days.

JOBC! was completed in 21 days against the delivery date of 25 days therefore no lateness.

JOB D was finished in 29 days against a promise delivery of 20 days. There's lateness.

JOB E: There was no lateness as Job was finished in 35 days against 40 days promised delivery

Jobs	Process Time	Flow Time	Due	
A	9	9	16	0
B	7	16	20	0
C	5	21	25	0
D	8	29	20	9
E	6	35	40	0

i) Total completion time = 35 days.

ii) Average completion time = Total flow time ÷ No of Job: $\frac{100}{5} = 22$ days

iii) Average numbers of work at the center = Total flow time divided total process time.

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100/35 = 3.14 days

iv) Average Job lateness = Total lateness / no of Jobs = 9/5 = 1.8 days

v) Late Job = Job B // <Note late Job is called "Tardy Job". i.e the late Job is "1".

3/01/22:

Continuation of Linear Programming (Faculty seen)

Linear programming problem formu-lation

B×G produces both interior and exterior paints from two raw materials, M_1 and M_2. The following tables provides the basis data of the problem

Tons of Raw material per ton of	Exterior paint	Interior paint	Maximum daily availability (ton)
			Interior paints.
Raw material 1	6	4	24
Raw Material 2	1	2	10
profit per ton (₦,000)	5	4	

Market survey restrict the maximum daily demands of interior paints to 4(tons). Additionally, the daily demand for interior paints cannot exceed that of exterior paints by more than 1 ton

B×G would like to determine the optimum product means of interior and exterior paints that maximizes the total daily profit. formulate the problem as a linear program.

solution:

step1: Define the decision variables.

let X_j (J = E, I) be the number of tons product J produced daily.

Note: No of interior & exterior paints to be produced is going to be denoted X_E and X_I

objective function:

Maximize; $Z = 5000X_E + 4000X_I$

Note: we are to maximize based on the available resources.

for material one (1):

Subject to; $6X_E + 4X_I \leq 24$ (material 1)

$1X_E + 2X_I \leq 10$ {material 2

$X_I \leq 4$ <market restriction

$X_E \leq 1 + X_I$

$X_E - X_I \leq 1$

problem 2: B×G produces three products by using three raw-materials and two processes. Denote the products by P_1, P_2, P_3 and the raw materials by M_1, M_2, M_3 while

ocesses are denoted by P_1, P_2, P_3.
ach unit of product 1, 2, and 3 req
es two (2kg), 4kg and 6kg of
aterial 1 respectively. furthermore,
ach units of product 1, 2 and 3 is
ade from 3kg, 5kg and 7kg of
aterial 2 respectively. While each
its of product 1, 2 and 3, requires
kg, 2kg and 4kg of material 3 res
pectively. The processing time of a
it of each of three products of
ocess 1 is 4hrs, 5hrs and 7hrs
spectively, also each of the three
oducts requires 1 2hrs, 3hrs & 5hrs
processes
product 2 respectively. The monthly
ailability of the three materials is
ited to 120kg, 100kg and 150kg
spectively. Monthly processing time
rough the two processes are 240hrs
nd 270hrs respectively. In addition
mand forecast have shown that not
re that 35 units of product two
ould be produced monthly. The
profit
ofit contribution of a unit of each
the three products is ₦55,
75 and ₦80 respectively.
rmulate the problem as a linear
rogram.

Solution.

Step 1: Define the decision variable.
Let X_j ($J=1,2,3$) be the number of units of product j to produce monthly

X_1 = No of product 1, to be produced monthly
X_2 = No of product 2 to be produced monthly
X_3 = No of product 3 to be produced monthly

Objective function: (using the profits of each products contribution.

Max: $Z = 55X_1 + 75X_2 + 80X_3$ (maximize (Z) profit. constant

Note: max this function is subject to some

subject to: $2X_1 + 4X_2 + 6X_3 \leq 120$ M_1
$3X_1 + 5X_2 + 7X_3 \leq 100$ M_2
$X_1 + 2X_2 + 4X_3 \leq 150$ M_3

for d processes products:

$4X_1 + 5X_2 + 7X_3 \leq 240 \rightarrow P_1$
$2X_1 + 3X_2 + 5X_3 \leq 270 \rightarrow P_2$

Note: $X_2 \leq 35$

$\therefore X_1, X_2, X_3 \geq 0$

Note: We have 6 slack variables.
each constraint has it own slack variable.

Graphical solution of Linear P.
We can only use graphical method to solve for 2 variable. for more than 2 variable we use "Simplex method"

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Topic: Transportation Model

DR. Adebiyi

Transportation model is representing a system of goods from one points to another in the most economic manner. (these places are origin & destination).

This model is considered due to the 4 rights which:

I) Right quality: the product available most have the right quality for customers

II) Right quantity: The product must be available in the required quantity as demanded by the customer. and also ensuring the goods has good quality.

Note: Buying in qty helps customers to gain some relational powers.

III) Right time: It is not sufficient to be the right quality, quantity but at the right time i.e these three rights are essential.

IV) Right Manufacturing cost: These cost is of two components

I) cost of getting your Inputs into the organisation II) cost of moving the goods to the customers. These cost must be minimal (optimal) at all time.

Note: The origin are sources where the products are produced.

The destination are where the customers demands for the product.

Note: For a transportation model to be solved, it must be balanced.

Balance Transportation model: This exist when the total demand is equal to the total supply or vice versa. otherwise is unbalanced

Unbalanced T. model: This is a situation where the total demand is greater than the total supply or vice versa.

the supply is related to the "origin" while the demand is related to the destination.

Note: In other to make an Unbalanced T. model balanced, a dummy needs to be Introduced or created.

A Dummy: A dummy is a row or column created to absorb the difference btw the total demand & total supply with zero cost of transportation (0).

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Solution procedure to transportation model:

There are two solution procedures to transportation model:

i) Initial basic feasible solution

ii) Optimum solution

A) Initial basic feasible solution: Under this, there are three m

i) North West Corner Method ii) Least cost method

iii) Vogel Approximation method.

B) Optimum solution: Under this, there are two methods:

i) Stepping stone algorithm method ii) Modified distribution

Illustration 1: Bolurwatife Nigeria limited transport product from ies to four warehouses of the customers. The supply capacity of t factories, cost of transportion as well as the demands from the u uses are presented in the table below.

Demand / Supply	Destination D_1	D_2	D_3	D_4	Supply
S_1	190	300	500	100	700
S_2	700	300	400	600	900
S_3	400	80	700	200	1800
Demand	500	800	1700	1400	~~3400~~ 3400

Note: supply → origi
Demand Destin

You are Required: ① To formulate the problem as a linear programming m

ii) solve for Initial basic feasible solution.

iii) Calcute the total cost of each method.

Note: 3 by 4 matrix is an effectiveness matrix and it can only be eff

If the model is balanced. In a 3 by 4 matrix, we have 12 cells of cells in an effectiveness matrix.

houses two things which are

D cost of transportations from one units from a perticular source f

ii) house the quantity that is transported from one source to a destinati

Note: When nothing is transported, the quantity & cost will be zer

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Solution

Demand / ine	D_1	D_2	D_3	D_4	Supply
S_1	500 (a) 190	(b) 300 200	500	100	~~700~~ ~~200~~ 0
S_2	700	(c) 300 600	(d) 400 300	600	~~900~~ ~~300~~ 0
S_3	400	80	(e) 700 400	(f) 200 1400	~~1800~~ ~~1400~~ 0
mand	~~500~~ 0	~~800~~ ~~600~~ 0	~~700~~ ~~400~~ 0	~~1400~~ 0	3400 3400

Linear Programming: The four steps:

Identification of decision variable (either max or minimise)

) Statement of objective function

) Add the constraint (either structural or non-structural)

) Add the Non-negativity function

$190X_{S_1D_1} + 300X_{S_1D_2} + 500X_{S_1D_3} + 100X_{S_1D_4} +$

$700X_{S_2D_1} + 300X_{S_2D_2} + 400X_{S_2D_3} + 600X_{S_2D_4}$

$400X_{S_3D_1} + 80X_{S_3D_2} + 700X_{S_3D_3} + 200X_{S_3D_4}$

) To maximise profit, gain or benefit while

e minimise cost, threat, risk etc.

Minimize (Z): i.e minimising the cost of T.

) Subject to constraint; S.T.

$X_{S_1D_1} + X_{S_1D_2} + X_{S_1D_3} + X_{S_1D_4} = 700$ } Supply constraint

$S_2D_1 + X_{S_2D_2} + X_{S_2D_3} + X_{S_2D_4} = 900$

$S_3D_1 + X_{S_3D_2} + X_{S_3D_3} + X_{S_3D_4} = 1800$

$D_1S_1 + X_{D_1S_2} + X_{D_1S_3} = 500$ } Demand constraint

$D_2S_1 + X_{D_2S_2} + X_{D_2S_3} = 800$

$D_3S_1 + X_{D_3S_2} + X_{D_3S_3} = 700$

$D_4S_1 + X_{D_4S_2} + X_{D_4S_3} = 1400$

) Non-Negativity Constraint.

$X_{ij} \geq 0$ for all i,j.

i.e any unknown ~~or~~ quantity transported from ij can either be greater or equal to zero. Where ij are the rows & column.

) solve for Initial basic feasible solution: Since one method was specified, then you try the three method

(a) North West Corner Method

) partition or divide the effective matrix steps to four cardinal point.

) Locate the most north west cell:

) make an optimum allocation to the cells without violating the row and column constraint.

An optimum allocation is the minimum between the demand and the supply constraint.

) Repeat the process till your total allocation equate the total demand and total supply.

Allocation	Units	cost	Total cost
a	500	190	95,000
b	200	300	60,000
c	600	300	180,000
d	300	400	120,000
e	400	700	280,000
f	1400	200	280000

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Least cost method:

demand / supply	D_1	D_2	D_3	D_4	Supply
S_1	150	300	200	700 100	
S_2	200 700	300	700 400	1600	7000
S_3	300 450	900 600 150	200	700 200	900 200 0
Demand	~~500~~ 200	~~800~~ 0	~~700~~ 0	~~1400~~ 700	1800 1000 3400 700 0 / ~~3400~~ 3400

Note: 1) Consider the ~~effectiveness~~ matrics & locate the cell with the minimum cost

Step 2: make optimum allocation to that cell.

iii) Repeat till your total allocation equates total Demand & Total supply.

Note: for the to exist in T.M two or more cells must have a location & must be feasible for allocation.

Note: 300 unit cost does not have the. You break a tie. in T.M with a cell that has maximum allocation.

So allocate to 400 units cost with supply of 900 & demand of 700

Note: When the demand & supply are zero (0) then it should be noted that the cell with its unit cost is not feasible

A degeneracy problem occurs in a T.M when the row + column − 1 is not equal to the number of allocation in such situation, a ~~fake~~ fictious allocation will be made in order to make the row + column − 1 equal to number of allocation.

A degeneracy problem prevents two things which are

1) The calculation of the total cost of transportation

ii) such solution could not be improved ~~include~~ include upon.

Allocation	Units	Cost	Total cost
a	800	80	64,000
b	700	100	70000
c	700	200	140,000
d	700	400	280,000
e	300	400	120,000
f	200	700	140,000
			TC = ₦814,0

r + C − 1 = 3 + 4 − 1 = 6 //.

Note: The least cost method gave a less cost than the North west corner method, so the least cost method is better.

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D Vogel Approximation Method.

demand / upply	D_1	D_2	D_3	D_4	Supply	P_1	P_2	P_3
S_1	500 (b) 190	300	500	200 100	~~700~~ 200	10	90	400
S_2	740	300	700 (e) 400	200 (f) 600	900 200	100	200	200
S_3	460	800 (a) 80	700	1000 (c) 200	~~1800~~ 1000 800	20	200	200
emand	~~500~~ 0	~~500~~	~~700~~ 0	~~1400~~ 400 0	3400			
P_1	210 210	220 0	100, 100	100, 100	3400			

ogel method 2nd ~~1st~~ introduced penalty

tep1: Identify the two minimum cost alonge ow and column, find the difference and ecord it as penalty

tep II: Identify the highest penalty, row or olumn.

tep III: Identify the least cost cell alonge he highest penalty.

tep iv) make optimum allocation to the ell with violation of the total demand & supply.

tep v! Repeat the process Until you are eft with two by two matrix and llow the least cost method

Allocation	unit	cost	Total cost
a	800	80	64,000
b	1000	200	200,000
d	200	100	20,000
e	700	400	280,000
b	500	190	95,000
f	200	600	120,000

Gabriel.

DR G.0.0. 02/02/22.

Topic: Project Management.

Note! The language of any bus is "finance.

Project management is a Unique activity (ie an activity that stands alone and it is not a routine).

Project management is a unique activity that has a time frame, designed to have specific objective and no bureaucracy. It is not just construction

Examples of project:

i) Building of bookshops
ii) Building of bridges
iii) Building of roads.

Note! Project does not happen every year or on weekly basis.

Why project management is different from general operation.

General operation is an operation done on daily basis (routine) but project managent is done once in a limited time frame.

* They are precious for new product
* They require special needs.
* Not all projects are successful.

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Key metrics or Important factors of Project Management.

I) Time II) cost III) Performance Objective

5 key success factors of P. Management

1) Drop-down commitment i.e you have the commitment of everyone in the organisation including the M.D such that ~~one~~ nobody doubt your idea i.e the backing of the org.

2) Respected and capable project Manager Eg our class project manager is Rebecca.

3) ~~You~~ Enough time to plan! When you plan, you succeed when you don't plan, you fail. planning makes execution easy.

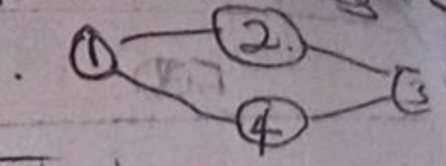
4) Careful tracking and control i.e monitoring

5) Good communication!

Main tools used in project management

1) Work break down structure: This means that all the activities involved in the project must be written down to ensure easy execution which is planning

2) Network diagram: That is you are connecting the activities by ~~arrows~~ nodes.



3) Gantt chart: This is a visual aid used to plan & monitor individual activities. eg the VC monitoring from his office if a project of building a hostle is going smoothly without going there physical. eg putting a call through

4) Risk management: for everything you do in life, there is a RISK eg Risk of been a student can occur due to strike, death etc this is the analysis of potential failure problem, assessment of their likelihood & consequences and Contigency Plans. Contigency plan means when risk occurs eg strike what will you do with that possible of strike, a good student might decide to learn how to drive within the three months strike

project Lifecycle.

Key decisions of project management

this implies all the things we need to do to have a good project.

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deciding which project to Implement
) Selecting the project manager i.e
oject manager selection who is respons
ble for recruitement, cost etc.
selecting the project team.
) planning and designing the project-
) managing and controlling the project
) Deciding if and when a project should be terminated

Project Life cycle.

here are four phases.

> Definition stage: That is you define which project you want to take
2) planning: listing all activities
3) Execution: This is Implementing the project (i.e project Implementation)
4) Termination: The conclusion of the project i.e bring the project to an end.

Question:

Assuming you have a network diagram with 6 activities and the duration to work. Required

I) Whats the length of each paths.

Path A = 1-2-4-5-6
= 8+6+3+1 = 18

Path B = 1-2-5-6
= 8+11+1 = 20

Path C = 1-3-5-6
= 4+9+1 = 14.

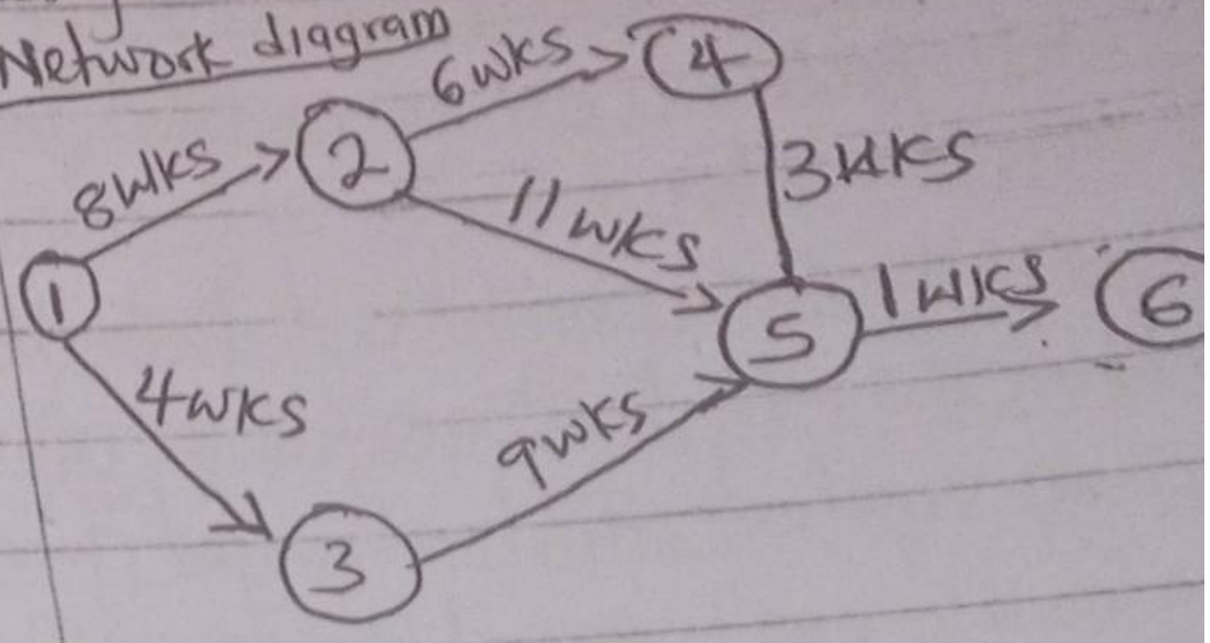
II) What is the Critical path? The Critical path is the longest path which is Past B. While the length of the cotrical path is 20%. The cotrical paths also shows the duration of the project. The duration of the project is 20wks.

III) Whats the slack: i.e the point where it is lagging behinde.

Slack = Critical path - path. each of the

project A: 20 - 18 = 2.
project B: 20 - 20 = 0
project C: 20 - 14 = 6.

Network diagram



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Dr Kayode: FBA 410 09/02/22.

Every system has it own peculiarity or activity that goes on in it
System is the mode of
MARKOV CHAIN PROCESS

this process enables us to have a better understanding about the behavioural pattern of customers. eg if a customer comes into the bank, they can either withdraw, deposit, borrow etc It is through this process we understand their behavioural pattern and be able to predict what they want to do at a particular time. It is applicable to all business environment

It is through this process we understand their behavioural pattern and to be able to predict their behavioural pattern to forecast sales.

Markov Chain Analysis: Many managem ent problems can be modelled using markov chain.

A markov chain is a probabilistic devices that is useful for represent -ing system that changes overtime eg employment, purchase of different brand of cars,

A markov analysis enables us predict the future state of any with the help of stage transiti probability analysis. In any a sequence of event or outcome independent of their preced or succeeding event.

Note: Every model has it own rule

Characteristics of a Markov c

i.e the conditions that must be -fied. There are considered:

i.) There are finite number of A state is the description of t behaviour pattern of the variables i system at a point in time. eg cars, companies, houses,

ii) States are both collectively exh -tive and mutually exclusive collectively exhaustive. All the sta you can be, can be known eg tossing of a die, all the situation can be known. tossing of dies mut mutually exclusive! The occurence of one, does not affect the occura of the other at the same time both cannot occur at once un Markov chain eg you cannot buy toyota today & buy honda also to

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iii) The transition probability depends on only the current state of the system. <i.e the next action of the variable does not depend on what happened in the past but the present situation.

iv) The long probability of been in a particular state will be constant overtime. The long run is the "steady state". → The steady state is just like a break-even state where no matter what you do, you will stil remain at the break-even. i.e the prob of moving from diff stages is constant.

v) The sum of transition probability of moving to alternative state in the next time period given a state in the current time period must be one.

Adelogo, a large retail grocery has been studying the brand switching or transition behaviour of each customers in the purchase of two major brands of beverages which are bournvita and milo.

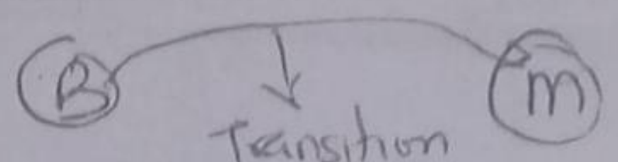
solution

*We have 2 state <bournvita & milo.

* We have 4 scenario that can take place. (B, M, Note → Transition)

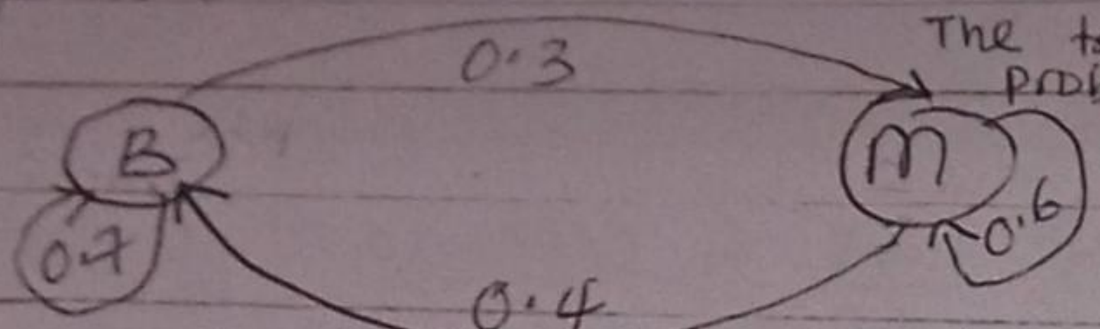
The marketing staff of Adelogo have collected data over the past year to support the following behavioural pattern. there is a 70% likelihood that a buyer of bournvita week will purchase bournvita again the following week & a 30% likelihood that the customer will switch to milo. A customer who purchase milo in 1 week will again purchase milo the following week with a 60% likelihood & will switch to bournvita with a 40% likelihood. Adelogo will like to use the result of this marketing research to deduce some generalisation concerning buyer behaviour in the future.

Transition: is when a customer moves from a state during one week to a state in another week we say a transition has taken place. sometimes it is referred to as "state transition".



The movement with the arrow is the transition.

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The transition probability diagram.

We use the matrix mostly when ≥ states are many in the situation.

We use the Transition Matrix to solve.

	B	M
B	0·7	0·3
M	0·4	0·6

Classifaction of states.

There are four classifaction of states:

) The state of the Markov chain can be classified based on the transition probability ~~Pij~~ Pij

*) A state j is ~~A~~absorbing if it returns to itself ~~the~~ with certainty in ones transition i.e $P_{ij} = 1$ (moving from bournvita to bournvita).

B) A state j is transient if it can' reach another state but ~~not~~ cannot if it -self be reached back from another state.

eg A customer can move from milo to bournvita but ~~from~~ Bournvita cannot go back to milo. eg $\lim_{n \to \infty} P_{ij}^{(n)} = 0$ for all i

) A state j is recurrent if the probability of been revisited from another state is 1. this can happen if and only if the state NOT TRANSIENT.

Pij

D A state ~~j~~ is periodic with t (time) $t > 1$ If a return is poss -ble only in $t, 2t, 3t$ --- st i.e there is a likelihood of coming back to your origin eg If Biso moves from her origin to others, she can still move back to her origin either in the 5th or step or 5 or more.

Note! The question we are solving is " Recurrent and Periodic ".

Note! Aperiodic → means wh there is no likelihood of custom to return to its origin

Step transition.

Note! The probability transition mus be equal to 1. (The rows mu be equal to one (1) eg $0.7 + 0.3 = 1$, $0.4 + 0.6 = 1$

Note: Number of row must

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next matrice.

(m)×n × m×(n)

must be equal.

	B	M	
B	0.7	0.3	P_1
M	0.4	0.6	

$$\begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} \times \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad P_2$$

$P^1 \times P^1 = P^{1+1} = P^2$

This indicate moving from B to B in the first week if I want to move to M it will take place in the 2nd wk then I have to multiply the original matrice by it self = $P^1 \times P^1 = P^{1+1} = P^2$ to get the values.

$a = (0.7 \times 0.7 + 0.4 \times 0.3) = 0.61$

$b = (0.7 \times 0.3 + 0.3 \times 0.6) = 0.39$

$c = (0.4 \times 0.7 + 0.6 \times 0.4) = 0.52$

$d = (0.4 \times 0.3) + 0.6 \times 0.6) = 0.48$

$$P_2 = \begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix}$$

In three weeks time, the transition will be. $P^2 \times P^1 = P^3$.

$$\begin{bmatrix} 0.61 & 0.39 \\ 0.52 & 0.48 \end{bmatrix} \times \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$$

$P_2 \quad P_3 \quad P_1$

$$= \begin{bmatrix} 0.583 & 0.417 \\ 0.556 & 0.444 \end{bmatrix}$$

In the long run we use Π_i to represent it.

$(\Pi_B, \Pi_m) = (\Pi_B \Pi_m) \begin{bmatrix} 0.7 & 0.3 \\ 0.4 & 0.6 \end{bmatrix}$

$\Pi_B = 0.7\Pi_B + 0.4\Pi_m$ — eqn (1)

$\Pi_m = 0.3\Pi_B + 0.6\Pi_m$ — (2)

$\Pi_B + \Pi_m = 1$ — (3)

use any of the equation but must use equt (3) - using eqn (1)

$\Pi_B = 0.7\Pi_B + 0.4\Pi_m$ — (1)

$\Pi_B + \Pi_m = 1$ — (3)

$\Pi_B = 1 - \Pi_m$.

$\Pi_B = 0.7\Pi_B + 0.4\Pi_m$.

$P^N = P^{N-1}(\Pi_B \Pi_m)$

$1 - \Pi_m = 0.7(1 - \Pi_m) + 0.4\Pi_m$

$1 - \Pi_m = 0.7 - 0.7\Pi_m + 0.4\Pi_m$.

$1 - 0.7 = -0.7\Pi_m + 0.4\Pi_m + \Pi_m$.

$0.3 = -0.3\Pi_m + \Pi_m$.

$0.3 = +0.7\Pi_m$.

$\Pi_m = \frac{0.3}{0.7} = 0.4286$.

Note: If we are given have 1000 custom then we multiply $1000 \times 0.3/0.7$ to kn the quantity of milo that w be bought in the long run.

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If $\Pi_m = 0.3/0.7$

Recall

$\Pi_B = 1 - \Pi_m$

$\Pi_B = 1 - 0.3/0.7$

$\Pi_B = \frac{0.4}{0.7}$ or $4/7$

To find $P^N = P^N = P^{n-1}$ (TPM)

where TPM = Transition of probability Matrix:

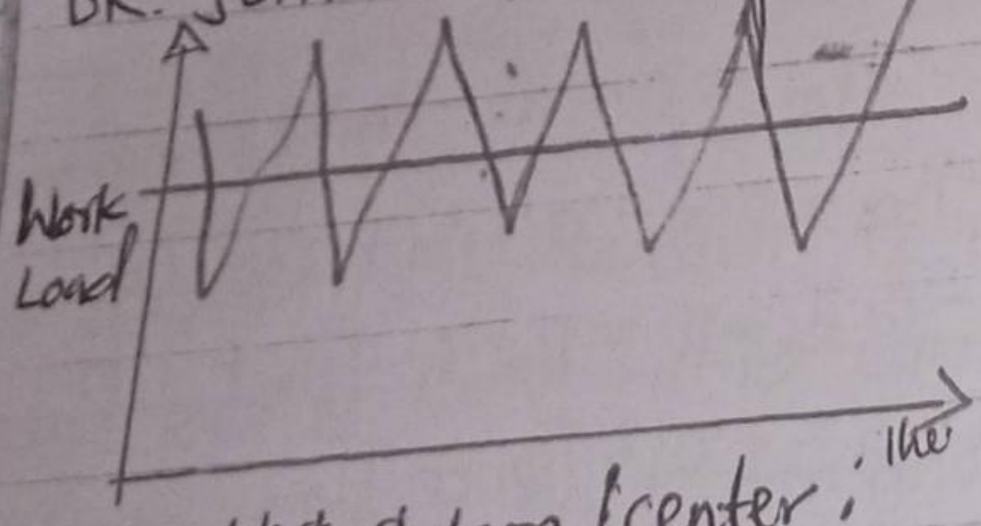
so for the 2nd step/week)

$P^2 = P^{2-1}$ (TPM)

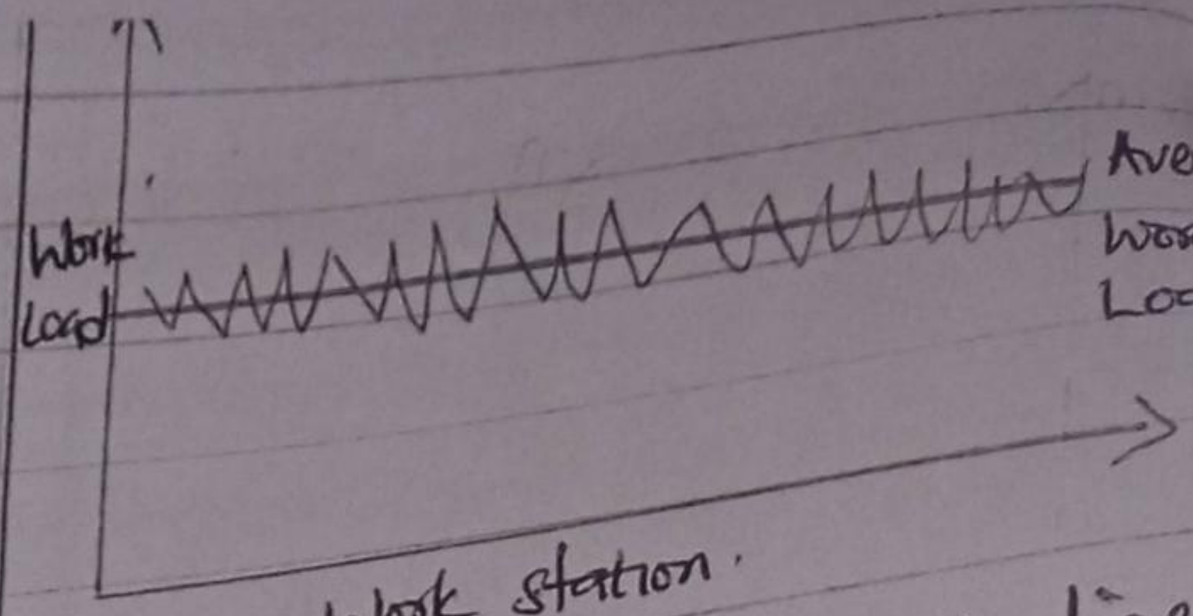
$P^1 \times (TPM)$ //.

Topic: Assembly Line Balancing

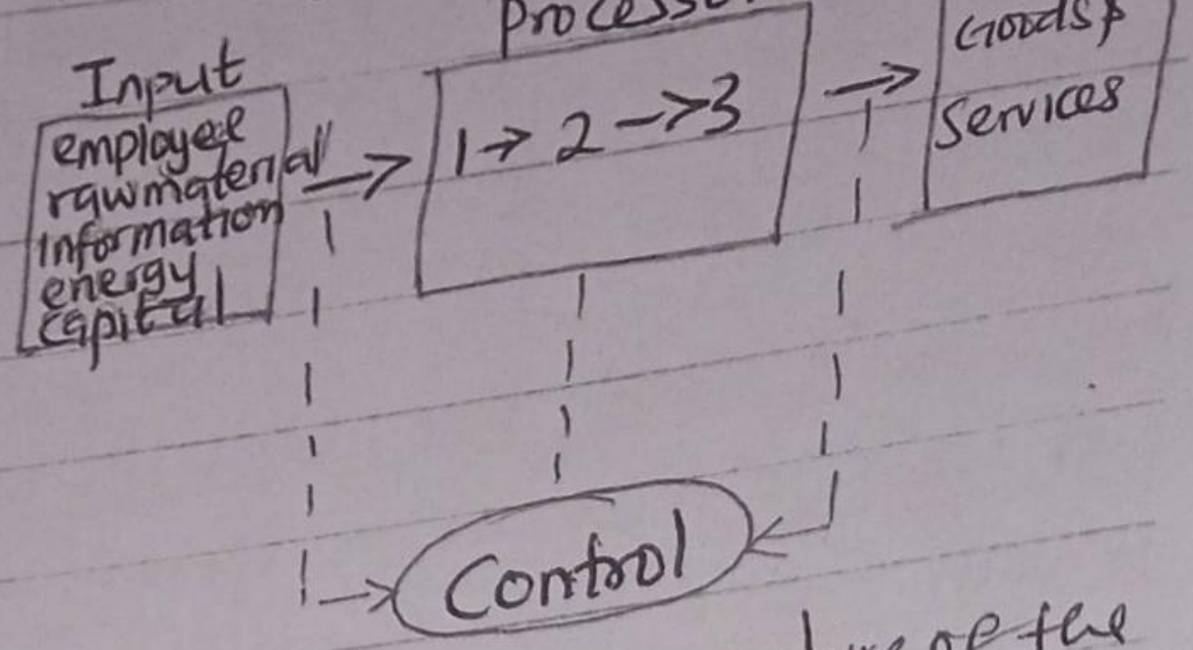
DR: Jonathan Ehimen Ekpudu



An Unbalance Assembly Line:



A fairly Balance Assembly line



Note: Depending on the nature of the goods produced, some may have 2 stages of processor before be-ming an output finished goods while some 3 stages.

Balance Assembly line: same time as begiven at the processor stage such that no stage is idle.

Unbalance Assembly line: Diff- f time/ hours are allocated, which causes idleness to st- stages. However a model h to be introduced to solve problem.

Work station are the sub stations where works are expected to be done on the assembly line.

objective of A-line is to balance & unbalance the work process using some models. This is to avoid idleness and increase efficiency.

Note: There is no perfect assembly line we can only have fairly balanced assembly line.

Note: The essence of Assembly line is to ensure efficiency.

Example I The table below shows the precedent relationship of the various task of an assembly and the desired output is 30 units, working hour per day is 8 hours

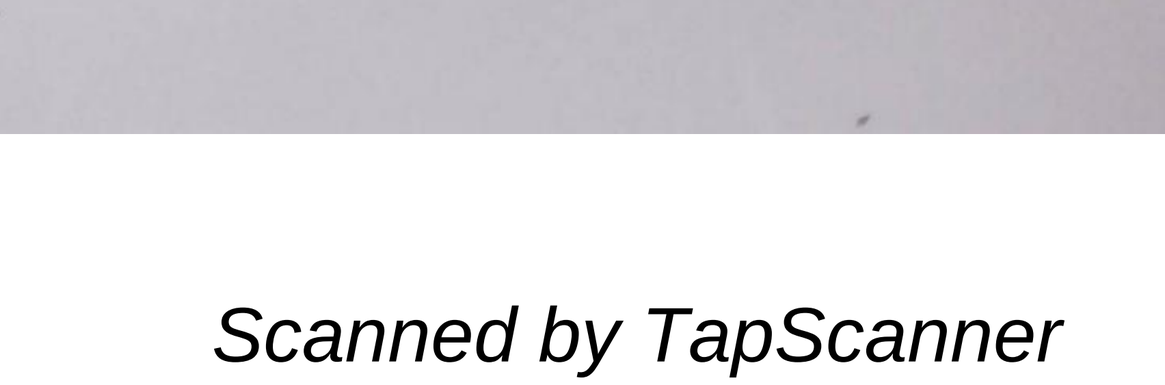
Assembly tasks (i)	Time mins.	preceding
A	9	—
B	5	—
C	4	—
D	5	A, B
E	6	D
	5	E, C
	7	F
	7	G

required
1) Construct the precedence activity Network diagram of the tasks.
2) select an appropriate heuristic method for allocating the task.
3) compute the cycle time.
4) Compute the ideal numbers of work stations.
5) Calculated total activities time and total idle time.
6) Compute the balance delays (BD) of the line (efficiency of the line).

Note: The balance delay is the idleness of the line usually in percentage which the opposite is the efficiency of the line.

Solution

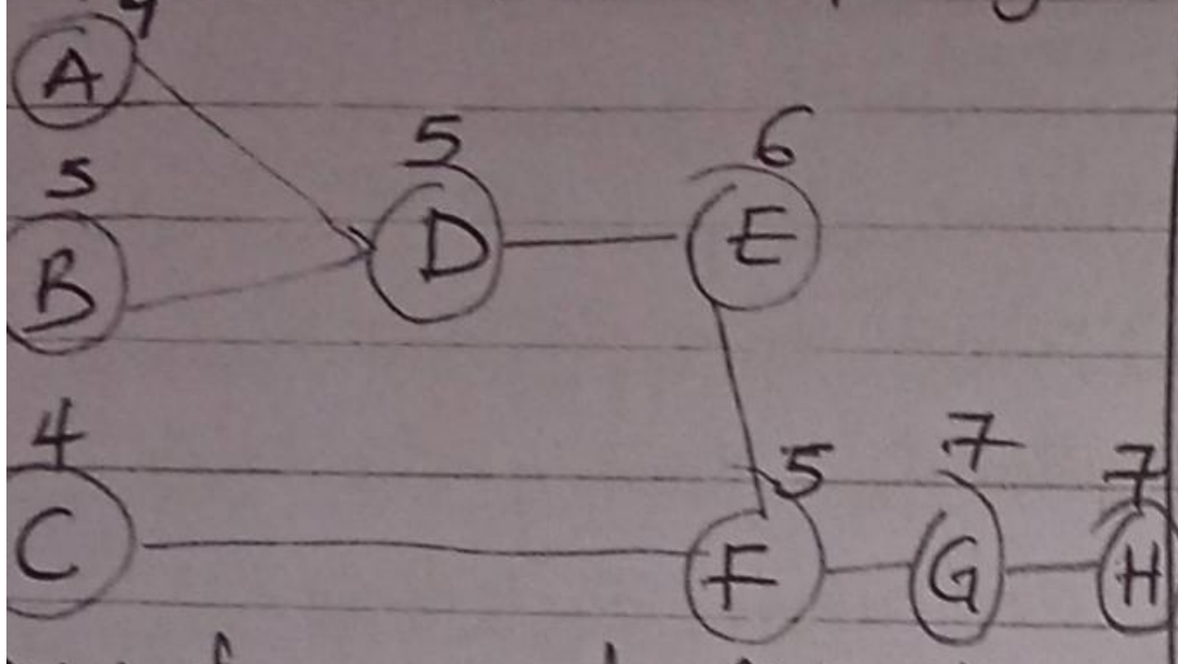
Note; Under Assembly line the nod are the activities while in project management the arro are the activities.



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Solution.

Precedent relationship diagram.



2) Western and Kilbridge method.

3) Cycle time is the maximum time that is allowed, if the assembly time must be balance.

Cycle time (C)

$C = \frac{\text{Available working time per day}}{\text{Desired output per day}}$

$C = \frac{8 \times 60 \text{ minutes}}{30} = 16 \text{ mins}$ //.

Note: since the we have ~~8~~ hrs & we were given our table in minutes then you first convert the 30 hrs to minutes. = 8 × 60 mins = 480 mins. Therefore, if the, work center must be balanced, we must spend 16 minutes on each work station.

4) Idle numbers of work station (N

$N = \frac{\sum ti}{C} = \frac{48}{16} = 3$ //.

Note: The idle time is 3 work station not 8.

eg Assuming we got 3.42, or 3.002 we have to round up to 4.

Note $\sum t_i$ = is the summation of the time in minutes. while

$C \rightarrow$ is the Cycle Time.

After calculating the ~~idle time~~ work station, we got 3, we have to balance our table.

Work station	Task	Time	Idle Time
1	A, B	14	2
2	C, D,	15	1
3	F, G	12	4
		41	7

These are the remainder since the bench mark is

Note: H, will be regarded as 16mins unnecessary station that must be discarded.

5) Total Activity Time is = 41
Total Idle Time is = 7.

6) Balance delay is the inefficiency of the line is the that is idle time since the

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Efficiency is the total time (41) (7) Inefficiency

b) Balance delay: 3(16) (41)

$$BD = \frac{NC - \sum ti}{NC} \times 100\%$$

$$NC(3(16))$$

NC = Work station x cycle time.

BD = The balance delay

cycle time

The efficiency in number is 41

Note: while the efficiency in percentage is $\frac{41}{48} \times 100$

The inefficiency in number is 7.

while the inefficiency in percentage is $7/48 \times 100$ %.

DR OYENUGA LEKAN. FBA 414.

IC: THEORY OF GAMES.

Strategies

Player B ⟨ minimize loss

	B_1	B_2	B_3	min gain
A_1	q_{11}	q_{12}	q_{13}	q_{11}
A_2	q_{21}	q_{22}	q_{23}	q_{21}
A_3	q_{31}	q_{32}	q_{33}	q_{33}

(q_{31} q_{22} q_{13}) least

max

* Assume the two players are rational
* A wants to maximise gain while B wants to minimize loss.
* The have finite number of strategy they have to take for player B (B_1, B_2 and B_3). while player A strategy is (A_1, A_2, B_2).
* Every positive variable is a gain to A and loss to B and any negative variable is a gain to B and loss to A.

Two Types of Strategy:

* Pure Strategy: This is used when a player continues to use a particular strategy irrespective of the strategy the opponent uses.
* Two Principles under Pure Strategy
* Maximin principle
* Minimax

Maximin Principle: We look at the minimum gain, then after which we consider the maximum gain, and the choose the "max min".

* Mixed Strategy.

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Illustration:

Saddle point = $ the value of the gain of the game

Company

		B_1	B_2	B_3	B_4	Min	Max
Union	A_1	20	15	(12)	35	12	12.
	A_2	25	14	8	10	8	
	A_3	40	12	10	5	2	
	A_4	-5	4	11	0	-5	
	Max	40	15	12	35		
Max loss	Min			12.			

Strategy: is the different cost of action to be taken.

We want to obtain "Optimal Strategy"

* value of the game * Saddle point.

we have to know if the game is *fair, *strictly determinable: This is possible when the value of the maxmin must be equal to the minimax value.

Saddle point: This is the equilibrium point of the maximum gain & minimum loss. The value at the saddle point is the "value of the game"

Strictly Determinable = Maxmin = Value of the game = mini max

Fair! This is when the value of the game is zero (0) i.e no gain and no loss.

Does the question have a saddle point ? Yes. ; optimal strategy = A_1, B_3.

Note: If there is no saddle point then we conclude that the gain of the game is 0 for the gain & loss, & no method can be used.

*Two person Non Zero sum G
This is when the addition of maxmin and minimax does equate to zero.

*Two person zero sum game
This is when the addition of the maxmin and minimax equate to zero (i.e the gain + the loss = 0.
eg +12 - 12 = 0"

Illustration 2.

Two Person Non zero sum G

	B_1	B_2	B_3	B_4	M
A_1	-2	0	0	5	
A_2	4	2	1	3	
A_3	-4	-3	0	-2	
A_4	5	3	4	2	
max	5	3	4	5	
min			1		

mini max

A_2, B_3

value of game = 1 = 1 = 1 → strategy dot

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Since "maxmin & the minimax did not equate to zero ($1,+1 \neq 0$) then it is Two person Non zero sum game.

Illustration 3:- When you see a question with alphabet, neglect the alp

	B_1	B_2	B_3	min	max
A_1	2	4	5	2	
A_2	10	7	P	7	7
A_3	4	2	6	4	
max	10	7	6		
min			6		

optimal
The strategy is A_2 & B_3.
The game is not fair, & is not strictly determinable.
value of the game is in between 6&7.
$6 < V < 7$.

* Mixed Strategy: When we have a two person game without a saddle point, the decision taken will be based on probability such that it will be more of one part & less of the other

		Q_1 B_1	Q_2 B_2	min	max
P_1	H	8 q_{11}	[-3] q_{12}	-3	-3
P_2	T	[-3]	(1)	-3	
max		8	1		
min				1	-3

Under the mixed strategy, once there is no saddle point. Under 2 by 2

$$V = \frac{q_{11}q_{22} - q_{12}q_{21}}{q_{11} + q_{22} - (q_{12} + q_{21})}$$

$$= \frac{8(1) - (-3)(-3)}{15} = \frac{-1}{15}.$$

$$P_1 = \frac{q_{22} - q_{21}}{q_{11} + q_{22} - (q_{12} + q_{21})}$$

$$= \frac{1 - (-3)}{8 + 1 - (-3 - 3)} = \frac{4}{15}.$$

$$P_2 = 1 - P_1 \quad \therefore 1 - \frac{4}{15} = \frac{11}{15}.$$

$$Q_1 = \frac{q_{22} - q_{12}}{q_{11} + q_{22} - (q_{12} + q_{21})}$$

$$= \frac{1 - (-3)}{15} = \frac{4}{15}$$

$$Q_2 = 1 - Q_1$$

$$= 1 - 4/15 = 11/15$$

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S_A S_B

$\left[\frac{4}{15}, \frac{11}{15}\right]$ $\left[\frac{4}{15}, \frac{11}{15}\right)$.

$S_A \rightarrow$ The probability of player A, taking a strategy for $P_1 = 4/15$, while $P_2 = 11/15$

S_B = The probability of player B, taking a strategy for $\Phi_1 = 4/15$, while $\Phi_2 = 4/15$

Using Algebra method.

$a_{11} P_1 + a_{21} P_2 = V$.

$a_{12} P_1 + a_{22} P_2 = V$.

$a_{11} P_1 + a_{21} P_2 = a_{12} P_1 + a_{22} P_2$

$8 P_1 + -3 P_2 = -3 P_1 + 1 P_2$.

Recall under probability

$P_1 = 1 - P_2$.

$8(1-P_2) - 3P_2 = -3(1-P_2) + 1P_2$.

$8 - 8P_2 - 3P_2 = -3 + 3P_2 + 1P_2$.

$-8P_2 - 3P_2 - 3P_2 - P_2 = -3 - 8$

$-15P_2 = -11$.

$P_2 = \frac{-11}{-15} = 11/15$.

$\therefore P_1 = 1 - P_2 = 1 - 11/15$

$= 4/15$.

Value of the game: $a_{11} P_1 + a_{21} P_2 = V$

$8(4/15) + -3(11/15) = V$

$\frac{32}{15} - \frac{33}{15} = V \quad \therefore V = \frac{-1}{15}$.

solve for Φ_1.

solving for Φ_1 & Φ_2.

$a_{11} \Phi_1 + a_{21} \Phi_2 = V$.

$a_{12} \Phi_1 + a_{22} \Phi_2 = V$

$a_{11} \Phi_1 + a_{21} \Phi_2 = a_{12} \Phi_1 + a_{22} \Phi_2$.

$8(\Phi_1) + (-3)\Phi_2 = -3\Phi_1 + 1\Phi_2$.

Note: $\Phi_1 = 1 - \Phi_2$.

$8(1-\Phi_2) - 3\Phi_2 = -3(1-\Phi_2) + 1\Phi_2$.

$8 - 8\Phi_2 - 3\Phi_2 = -3 + 3\Phi_2 + 1\Phi_2$

$-8\Phi_2 - 3\Phi_2 - 3\Phi_2 - \Phi_2 = -3 - 8$.

$-15\Phi_2 = -11$.

Divide both side by Φ_2.

$\frac{-15\Phi_2}{-15} = \frac{-11}{-15}$

$\Phi_2 = 11/15$.

$\therefore \Phi_1 = 1 - \Phi_2$

$\Phi_1 = 1 - 11/15$.

$\Phi_1 = \frac{15 - 11}{15} = \frac{4}{15}$.

$\therefore \Phi_1 = 4/15$.

Value of the game.

$a_{11} \Phi_1 + a_{21} \Phi_2 = V$.

$8(\frac{4}{15}) + (-3)(11/15) = V$.

$\frac{32}{15} - \frac{33}{15} \quad \therefore V = \frac{-1}{15}$.

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